

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.



Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

Features span demographics, purchase details, and shopping behavior — including discounts, promo codes, and subscription status.



Data Preparation in Python

```
1 import pandas as pd
2 import numpy as np
3
4 # 1. Load raw data
5 df = pd.read_csv('data/raw_sales.csv')
6 customers = pd.read_csv('data/customers.csv')
7
8 # 2. Initial inspection
9 print(f"Raw sales shape: {df.shape}")
10 print(f"Raw customers shape: {customers.shape}")
11
12 # 3. Standardize column names
13 df.columns = df.columns
14     .str.strip()
15     .str.lower()
16     .str.replace(r"[^a-z0-9_]+", '_', regex=True)
17     .str.strip('_')
18 )
19 customers.columns = (customers.columns
20     .str.strip().str.lower().str.replace(r"[^a-z0-9_]+", '_', regex=True)
21     .str.strip('_')
22 )
23
24 # 4. Data type cleaning
25 df['order_date'] = pd.to_datetime(df['order_date'], errors='coerce')
26 df['quantity'] = pd.to_numeric(df['quantity'], errors='coerce')
27 df['unit_price'] = pd.to_numeric(df['unit_price'],
28     errors='coerce')
29 df['customer_id'] = df['customer_id'].astype('Int64')
30
31 # 5. Handle missing values
32 df['quantity'] = df['quantity'].fillna(0)
33 df['unit_price'] = df['unit_price'].fillna(0.0)
34 df['discount'] = df['discount'].fillna(0.0)
35 df.dropna(subset=['order_date', 'customer_id'], inplace=True)
36 customers['email'] = customers['email'].fillna('unknown@example.com')
37
38 # 6. Clean text fields
39 customers['first_name'] = customers['first_name'].str.strip().str.title()
40 customers['last_name'] = customers['last_name'].str.strip().str.title()
41 customers['country'] = customers['country'].str.strip().str.title()
42
43 # 7. Remove duplicates
44 df = df.drop_duplicates()
45 customers = customers.drop_duplicates(subset=['customer_id'])
46
47 # 8. Business logic: total_amount
48 df['total_amount'] = (df['quantity'] * df['unit_price']) * (1 - df['discount'])
49
50 # 9. Final inspection
51 print(f"Clean sales shape: {df.shape}")
52 print(f"Clean customers shape: {customers.shape}")
53
54 # 10. Save cleaned data
55 df.to_parquet('outputs/clean_sales.parquet', index=False)
56 customers.to_parquet('outputs/clean_customers.parquet', index=False)
57 print("Data cleaning pipeline completed successfully.")
```

CHECKLIST

- ☒ TYPES
- ☒ MISSING
- ☒ DUPLICATES
- ☒ LOGIC
- ☒ EXPORT

GOOD DATA IS GOOD BUSINESS

→ Clean & Standardize

Imputed missing ratings by category median; renamed columns to snake_case

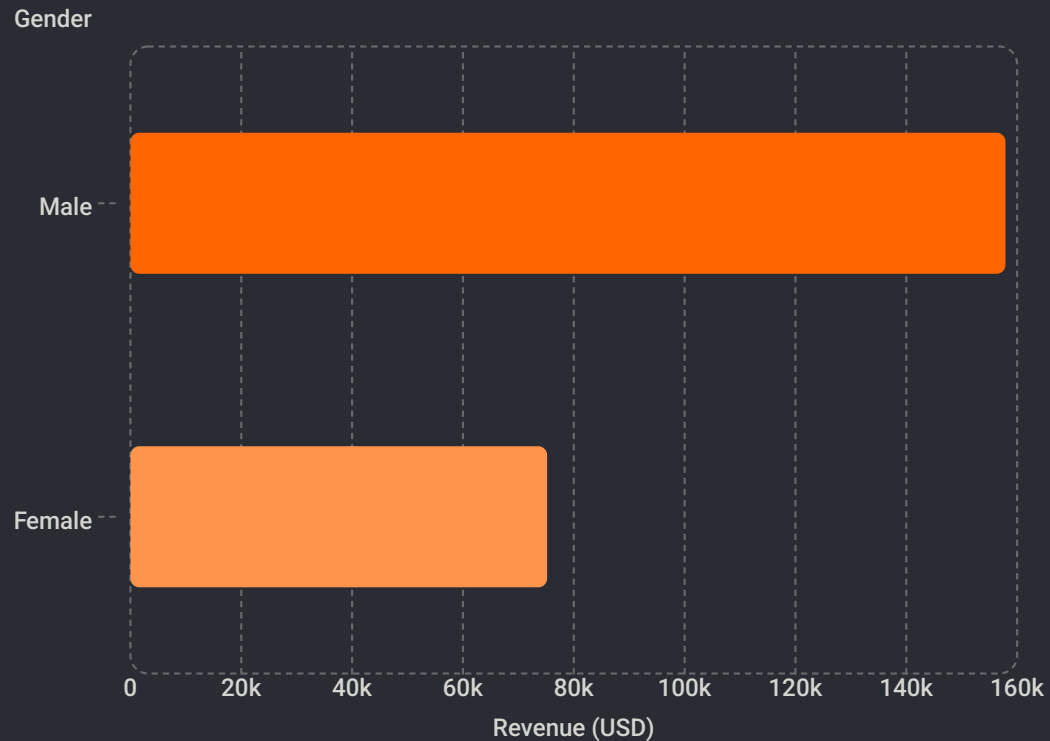
→ Feature Engineering

Created age_group and purchase_frequency_days

→ Load to PostgreSQL

Cleaned DataFrame loaded for SQL analysis

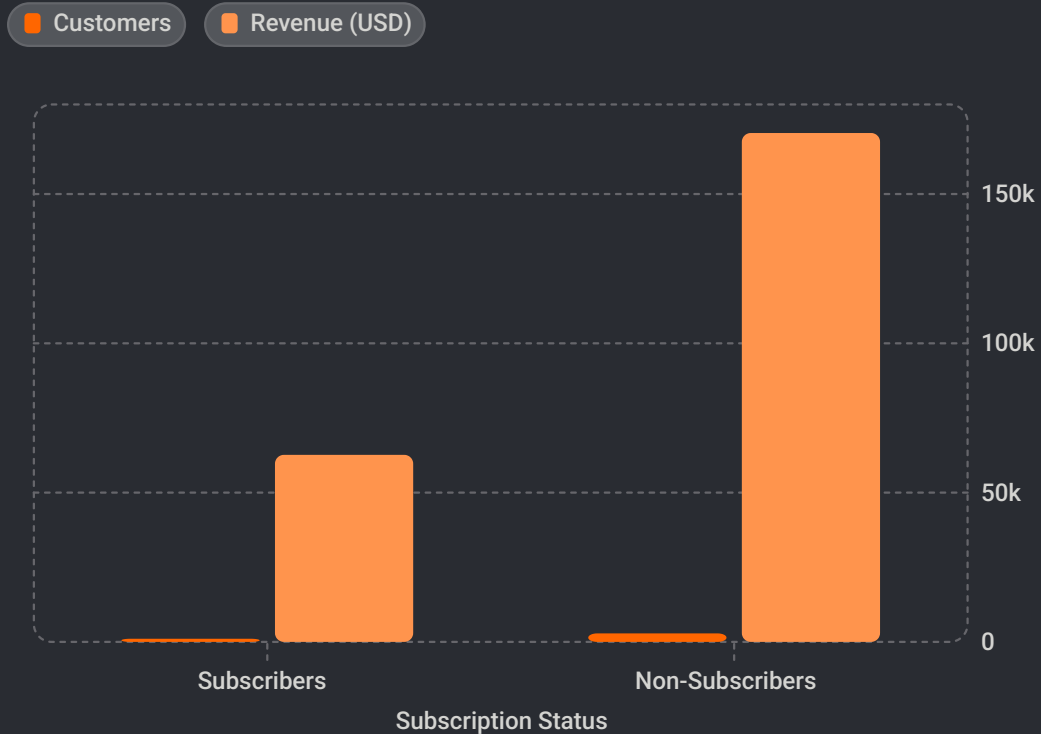
Revenue by Gender



Key Finding

Male customers generated **2.1× more revenue** than female customers — a critical segment to understand and target.

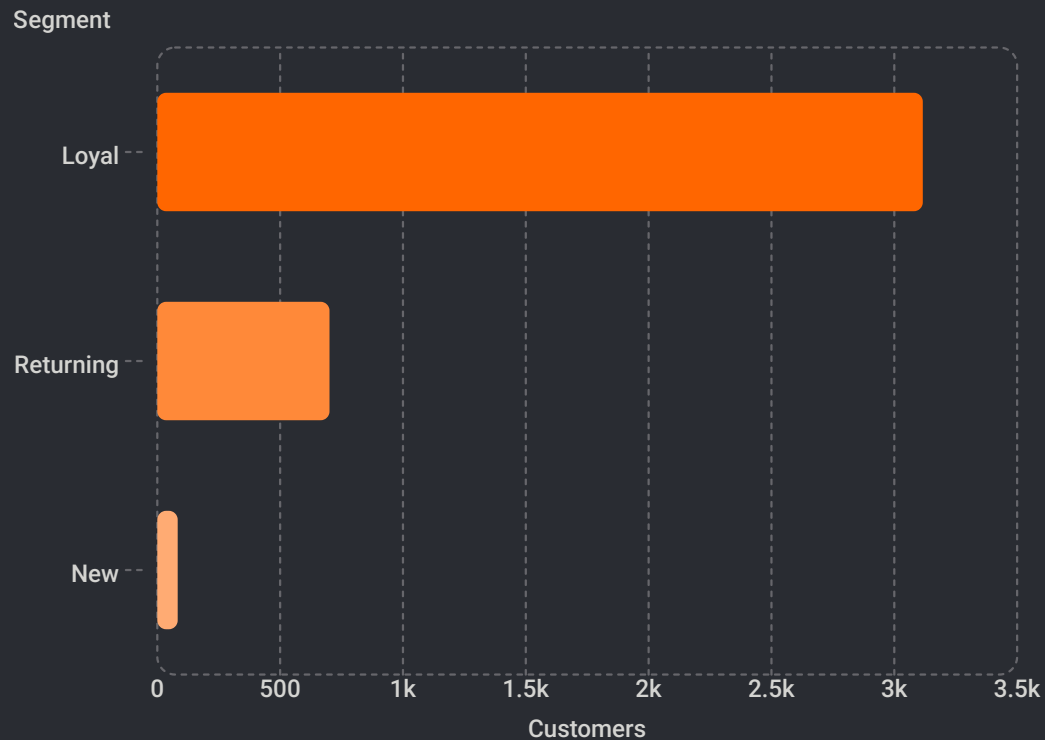
Subscribers vs. Non-Subscribers



Insight

Non-subscribers drive **73% of total revenue** despite similar average spend (~\$59). Subscription conversion is a major growth opportunity.

Customer Segmentation



Segment Breakdown

Loyal: 3,116 customers (10+ purchases)

Returning: 701 customers (2–10 purchases)

New: 83 customers (1 purchase)

80% of customers are already Loyal — retention programs can deepen engagement.

Top Products & Categories



Clothing

Blouse, Pants, Shirt — highest order volume



Accessories

Jewelry, Sunglasses, Belt lead the category



Footwear

Sandals, Shoes, Sneakers top sellers

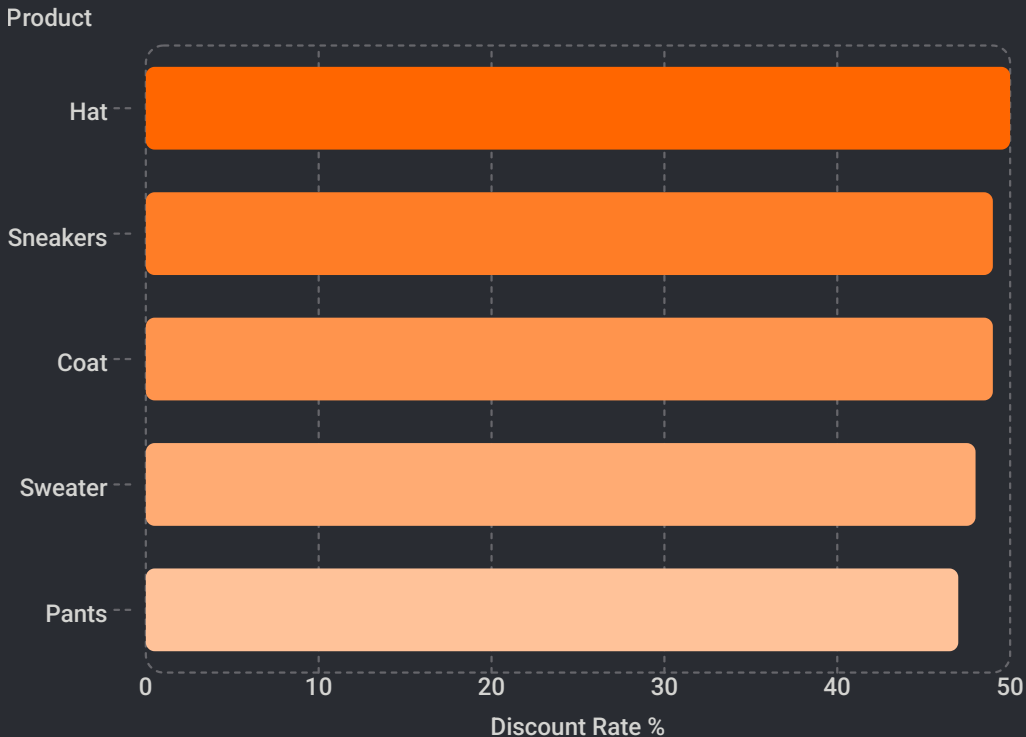


Outerwear

Jacket and Coat dominate purchases

Discounts & Ratings

Products Most Dependent on Discounts



Top-Rated Products

Gloves (3.86★), Sandals (3.84★), Boots (3.82★) lead in customer satisfaction.

High discount rates on Hats & Sneakers suggest price sensitivity — review margin impact.

Power BI Dashboard

3.9K

Total Customers

\$59.76

Avg. Purchase Amount

3.75★

Avg. Review Rating

27%

Subscription Rate

Interactive filters: Subscription Status, Gender, Category, Shipping Type.
Visuals cover revenue, sales, and age group breakdowns.

Made with **GAMMA**



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers



Review Discount Policy

Balance sales with margins



Product Positioning

Highlight top-rated products



Targeted Marketing

Focus on high-revenue age groups